

Character Display and Stable Records

Keep state, presentation, transport, and evidence separate

The display shows meaning; TP14 shows transport; the record preserves facts.

Lesson	014 — component	Time	about 45 minutes
Prerequisites	Lessons 001–013	Board	Arduino Mega 2560
Progress	D13 blink, welcome screen, values, fault, recovery	Output	LCD text plus optional stable records
Safety	E1; USB-powered low-current circuit only		

Safety checkpoint. Disconnect USB before wiring. Identify the exact LCD module and its pin order from primary documentation. Do not assume its backlight includes a resistor. If the backlight current limit or polarity is unknown, leave pins A and K disconnected. Stop for heat, odor, resets, unstable power, a contrast control that places a rail across the potentiometer wiper incorrectly, or any disagreement between the identified module and the literal table. Remove USB power to stop the experiment.

1 Mission: make the little screen tell a story

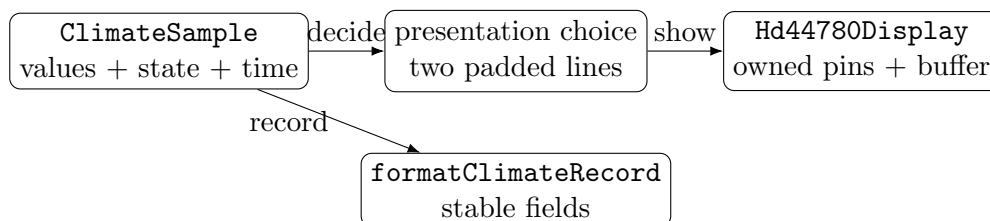
Present a deterministic climate snapshot without making the display the source of truth:

$$\text{ClimateSample} \rightarrow \begin{cases} \text{Hd44780Display} & \text{short human view} \\ \text{formatClimateRecord()} & \text{stable machine record} \end{cases}$$

The canonical sketch first displays a welcome screen, then replays four fixed samples. This isolates presentation from sensing: the same sample selects both LCD text and a record, while neither path modifies the sample. Lesson 015 will replace the fixed table with the scheduled sensor from lesson 013.

1. A short D13 blink says its own pin and all six LCD pins were acquired.
2. LCD `READY` rewards correct supply, contrast, and signal wiring.
3. Two changing values show the display is alive.
4. `SENSOR TIMEOUT` appears, then the next view recovers by itself.

2 The four boundaries



`CharacterDisplay` is the substitutable presentation boundary. It defines lifecycle, caller-driven `update(now)`, readiness, two-row `show()`, and a desired-line snapshot. `Hd44780Display` is the concrete 16-by-2, four-bit adapter. It owns RS, E, and D4–D7 through six fixed resource claims.

Construction is inert. `initialize()` validates distinct Mega digital pins, acquires all six before exposing behavior, configures them low, and starts a staged power-up sequence. A conflict or injected configuration failure releases every claim. Repeated initialization is inert.

`update(now)` advances at most one startup stage or one changed character. The caller supplies time; no background task or global callback exists. The bounded one-microsecond enable pulse is an electrical transport operation, not an application schedule. `shutdown()` commands display off when possible, drives owned pins low, releases them to high impedance, and is idempotent. Destruction repeats this safely.

3 The exact build

Quantity	Part	Requirement
1	Mega 2560	USB powered
1	independently identified LCD1602	parallel 1–16 / VSS–K header required
1	10 k Ω linear potentiometer	contrast control only
0	backlight connection	A/K stay open until the exact module rating is known
1	D13 indicator	built-in LED or external resistor-limited LED
1	breadboard and jumpers	route by LCD pin number, not wire color

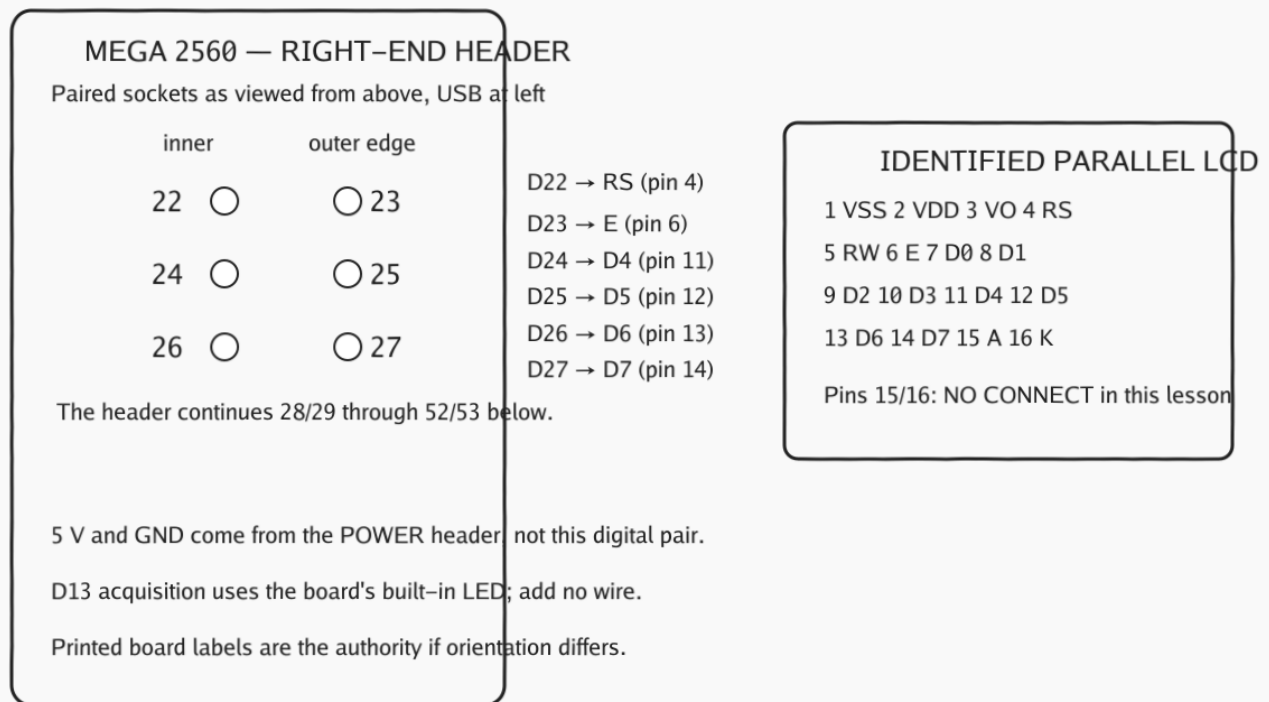
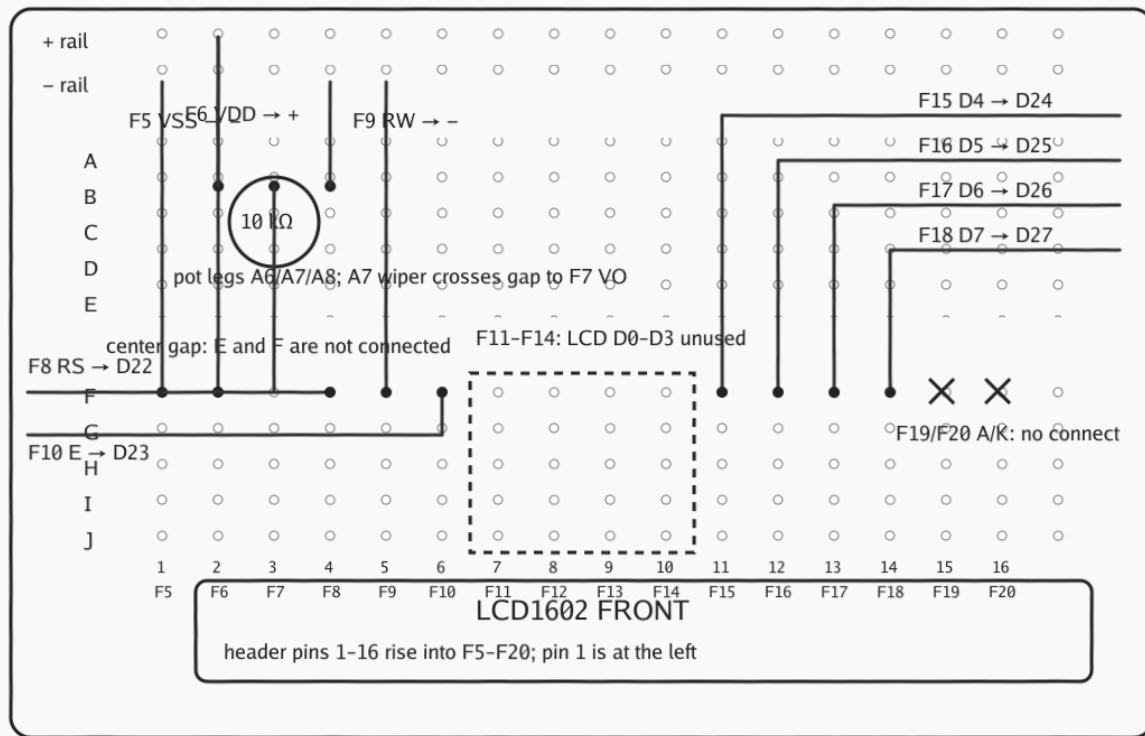


Figure 1: Pencil pin map. Follow the printed Mega socket numbers and the LCD1602 pin numbers. Alternative text: D22 through D27 map respectively to LCD pins 4, 6, 11, 12, 13, and 14.

BREADBOARD COORDINATES — LCD HEADER IN ROW F



Rails are separate: jumper + to Mega 5 V and – to Mega GND. Wire colors carry no meaning.

Figure 2: Breadboard close-up with the LCD facing you and pin 1 at the left. Alternative text: LCD supply and RW reach the rails, VO reaches the potentiometer wiper, six signals reach the Mega, D0–D3 remain empty, and backlight pins A and K remain disconnected.

4 Literal wiring

Function	Mega connection	Literal path
LCD supply	5 V and GND	identified VDD to 5 V; VSS to GND; never use Vin
contrast	VO	10 kΩ ends to 5 V/GND; wiper to identified VO
write policy	RW	identified RW directly to GND
register select	D22	D22 to identified RS
enable / TP14	D23	D23 to identified E; label this node TP14
data bit 4	D24	D24 to identified D4
data bit 5	D25	D25 to identified D5
data bit 6	D26	D26 to identified D6
data bit 7	D27	D27 to identified D7
unused data	none	D0–D3 remain unconnected in four-bit mode
backlight	pins 15 A and 16 K	leave disconnected in this lesson
acquisition	D13	use the Mega's built-in LED; no extra wiring

Inspect every path with USB removed. Set the contrast control near its midpoint before power. This lesson drives only the LCD logic supply. It makes no backlight-current assumption: pins 15 and 16 stay open. The retail family name does not establish a controller or revision. Continue only after the actual part independently

shows a parallel 16-pin row whose labels run VSS, VDD, VO, RS, RW, E, D0–D7, A, K in that order. An I2C backpack, missing labels, or a different order is a different specimen.

5 Startup and incremental work

An HD44780 controller is not ready when the MCU resets. The adapter models that fact explicitly:

claim → 15 ms → 3, 3, 3, 2 nibbles → function, off, clear, entry, on → ready.

Calls made before `ready()` only advance startup when their caller time is due. Once ready, `show(row, text)` updates the desired 16-character line. It rejects a null pointer, nonprinting ASCII, an unknown row, or text longer than 16 characters without touching the LCD.

Each later `update(now)` compares desired and shown buffers and writes one changed cell. A loop therefore remains responsive while a whole line changes. The desired snapshot is stable until another `show()` call.

6 Build it as five quick wins

Keep USB unplugged while moving wires. Power up only to enjoy the next win, then unplug again.

1. **Rails and contrast.** Connect LCD pins 1, 2, 3, and 5. Turning the potentiometer may reveal dark character blocks; their absence does not diagnose the circuit.
2. **Control pair.** Add RS on D22 and E on D23. Label the E wire TP14 so it is easy to recognize later.
3. **Four-bit data.** Add LCD D4–D7 in order to Mega D24–D27. Pins D0–D3 remain empty. The first promised payoff is a short D13 blink followed by LCD READY.
4. **Watch it move.** The first and second sample views advance without clearing or flicker.
5. **Watch it recover.** The third view says `SENSOR TIMEOUT`; the fourth returns to values automatically. Press RESET whenever you want to replay the whole sequence.

THE SCREEN TELLS THE STORY — RESET REPLAYS IT

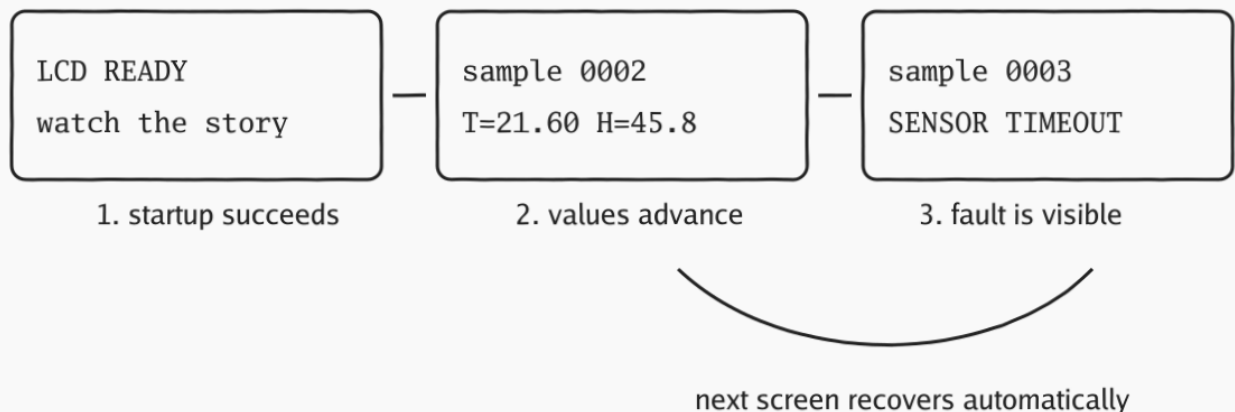


Figure 3: The visible reward sequence. Alternative text: LCD READY leads to changing climate values, then SENSOR TIMEOUT, followed by automatic recovery on the next screen.

```
void loop ()
{
```

```

const adk::TimePoint now (millis ());

display.update (now);

if (display.ready () && sampleDue (now))
{
    const adk::ClimateSample sample = decideSample (now);
    showSample (sample);
    recordSample (sample);
}
}

```

7 Stable records

The record format is ASCII, one line, with fixed field order and explicit units:

```
seq=42,state=valid,temp_centi_c=-125,rh_permille=456,at_ms=4294967290
```

Integers avoid decimal punctuation and floating-point formatting. Names encode units. States such as `transport_timeout`, `checksum_failure`, and `stale` remain distinct. The timestamp is the sample observation time, not the time the line happened to print.

`formatClimateRecord()` allocates nothing and does not use locale. The caller supplies a buffer. A too-small buffer is terminated and reports `CapacityExceeded`; a null or zero-capacity buffer reports `InvalidArgument`. Serial output in the canonical sketch merely carries this record. It is optional evidence: the LCD and TP14 provide the required non-Serial paths.

8 What each clue means

Claim	Prediction and observation	Interpretation
resource acquisition	D13 blinks for 250 ms after its own pin and all six LCD pins are acquired	no blink means setup did not complete
safe startup	before ready, signal pins start low and the LCD may remain blank	blank during the specified startup interval is not yet a sensor fault
transport	TP14 has bounded enable pulses as cells change	pulses prove attempted writes, not readable contrast or correct data wiring
presentation	fixed samples appear in the documented order	text order tests the presentation table and incremental flush
record identity	sequence increases once per displayed sample	one sample produced both views; locale cannot alter field punctuation
reset	D13 blinks and the welcome screen returns	the complete story is replayable without rewiring

Behind the scenes. Automated checks still exercise startup timing, rollback, bad text, resource conflicts, repeat shutdown, incremental updates, record boundaries, and transport failures. The learner gets the screen story instead of a test worksheet.

9 Diagnosis

Observation	Likely boundary	Next safe check
D13 remains off	claim or initialization failure	remove power; check distinct D22–D27 wiring and conflicts
D13 blink, no blocks or text	supply, contrast, or startup	remove power; verify VSS/VDD/VO and module identity
blocks but no text	RS/E/data mapping	remove power; compare every literal path
wrong characters	D4–D7 order or intermittent connection	remove power; continuity-check each data path
TP14 quiet while text queued	loop halted or transport failure	press RESET and watch the welcome sequence
TP14 active, unreadable LCD	contrast, supply, or data mapping	do not treat transport as presentation proof
record differs by locale	unsupported formatter introduced	return to the canonical sketch
old pixels after unplugging	LCD charge fading normally	wait for it to fade; reconnect and re-play

10 Make it yours

1. Replace the welcome screen with a family message of at most 16 printable characters per line.
2. Turn the four fixed views into a tiny weather-story animation.
3. Change one character at a time and watch how smoothly the adapter updates only the cells that changed.
4. Unplug USB during the timeout view, reconnect it, and watch the story restart cleanly from the welcome screen.

11 Next

Lesson 015 composes the scheduled climate sensor, this character display, stable records, and explicit health policy into an environmental station. It does not turn this learning circuit into a calibrated monitor or safety alarm.